

# Rajdeep Gill

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## Education

### University of Manitoba

Bachelor of Science in Computer Engineering

May 2026

4.46/4.5 GPA

## Experience

### Robinhood

Software Engineer Intern

May 2025 - Sept 2025

Toronto, ON

- Reduced customer margin calls by **24%** and boosted trading activity by **13%** through a **real-time** margin maintenance table built with React and TypeScript, featuring embedded trade execution and risk indicators.
- Architected a cross-platform margin experience using a **server-driven UI approach** to unify business logic across Web, iOS, and Android, driving a **43%** increase in margin usage.
- Redesigned the margin call experience utilizing server-driven UI and incorporating the maintenance table.
- Accelerated the adoption of a new **Protobuf** design system by implementing its foundational components in **Python and TypeScript**, and adding a **logging framework** to improve observability.

### Biosensors Research Lab

Machine Learning Research Intern

May 2024 - Dec 2024

Winnipeg, MB

- Built a custom **CNN** achieving **85% accuracy** on 10,000+ diffraction images, providing a scalable, **non-invasive alternative** to traditional staining methods for assessing cell health.
- Built a **model validation GUI** in Python displaying SHAP-based feature attributions, allowing researches to debug predictions cell by cell and improve model accuracy.
- Engineered an end-to-end **parallelized data processing pipeline** that improved the accuracy of morphological parameter extraction by **17%** and cut processing runtime by 75%.
- Deployed a **centralized MySQL database** to replace manual hard-drive sharing, eliminating data loss and versioning conflicts while enabling researchers to run **cross-experiment queries** and collaborate on shared datasets.

### Biosensors Research Lab

Machine Learning Research Intern

May 2023 - Aug 2023

Winnipeg, MB

- Engineered a **Python control framework** for lab oscilloscopes that abstracted low-level hardware commands into a high-level API, allowing **automation of entire experimental sequences** and cutting manual setup time.
- Slashed data analysis runtime by over **90%** (1 hour to <6 minutes) by refactoring legacy Python scripts, replacing inefficient loops with vectorized NumPy operations
- Developed a classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health on an unsupervised basis.

## Projects

### Void - AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

### Collaborative Document Editor (React, TypeScript, Socket.io, Flask)

- Engineered a collaborative text editor for up to 20 simultaneous users by broadcasting text operations, cursor locations, and online statuses, via WebSockets, achieving a sub-250ms synchronization latency
- Integrated Gemini to provide real-time suggestions for text, formatting and content generation, reducing writing time

### UM-Tracker (React, TypeScript, PostgreSQL)

- Created a course assignment tracker for the University of Manitoba, allowing students to link their current courses to keep track of their upcoming assignments and exams
- Implemented syncing with personal calendars to display a comprehensive view of the user's schedule

## Skills

**Programming Languages:** C++, Python, TypeScript, JavaScript, Java

**Frameworks and Libraries:** React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

**Tools:** Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf)